

MARIGOLD

Tagetes erecta (African Marigold) / *Tagetes patula* (French Marigold)

Quick Facts

Scientific Name	Tagetes erecta (African Marigold) / Tagetes patula (French Marigold)
Family	Asteraceae (Daisy family)
Common Names	Marigold, African Marigold, Genda (Hindi/Bengali), Zendu (Marathi), Banthige (Kannada), Chamanthi (Telugu), Marikkolundu (Tamil)
Native Origin	Mexico and Central America
Colors	Brilliant orange, yellow, golden yellow, lemon yellow, cream, and bi-coloured orange-yellow. Orange is the predominant colour in Indian commercial production.
Height	30 to 90 cm for African marigold (Tagetes erecta); 20 to 50 cm for French marigold (Tagetes patula).
Blooming Season	October to March is the main season across most of India. A second crop blooms June to August in some regions. Planted at the beginning of September for Navratri-Dussehra-Diwali demand, and again in November for the wedding season. Short-duration varieties can be brought to flower within 45 to 60 days from transplanting.

Distribution in India

The single most important commercial flower crop in India. Cultivated in virtually every state, with the highest production in Karnataka, Andhra Pradesh, Telangana, Tamil Nadu, and Maharashtra. West Bengal is a major producer for the Kolkata market. Karnataka alone accounts for approximately 30 percent of India's total marigold production. The Pune and Bengaluru wholesale markets are among Asia's largest marigold trading centres.

Description and Overview

Marigold (Tagetes) is India's most commercially important cut flower and loose flower crop. It is an annual herbaceous plant with a strong, distinctive aromatic scent arising from terpene compounds in the leaves and stems. The flower heads are composed of multiple layers of ray florets arranged in a globular or semi-globular shape. In *Tagetes erecta* (the African or American Marigold), the flower heads are 5 to 10 cm in diameter and produced on long stems. In *Tagetes patula* (the French Marigold), flower heads are smaller, 3 to 5 cm, and produced in greater abundance on compact plants.

Marigold has an unparalleled cultural significance in India. It is the cornerstone flower of Hindu religious festivals, weddings, and public celebrations. Marigold garlands are used to welcome guests, decorate deity idols, adorn doorways during festivals, and garland the deceased. The flower is deeply associated with Navratri, Dussehra, Diwali, and wedding ceremonies across all communities in India.

The economic scale of India's marigold industry is remarkable. The total annual production exceeds several million tonnes of fresh flowers, with a value running to thousands of crore rupees. Andhra Pradesh and Karnataka produce the bulk of the crop, which is then distributed through wholesale flower markets to every corner of the country. During festival peaks (Diwali, Dussehra, Ugadi, Pongal), the price per kilogram of marigold flowers can rise 10 to 20 times above the off-season price.

Beyond the domestic market, India exports marigold oleoresin (a concentrated extract of the yellow pigment lutein) and dried marigold petals to the European Union, United States, and East Asian countries. Lutein from marigold is used as a natural food colourant, in poultry feed to enhance egg yolk colour, and increasingly in nutraceutical supplements for eye health.

Blooming Season and Conditions

October to March is the main season across most of India. A second crop blooms June to August in some regions. Planted at the beginning of September for Navratri-Dussehra-Diwali demand, and again in November for the wedding season. Short-duration varieties can be brought to flower within 45 to 60 days from transplanting.

Cultivation

Marigold is propagated from seeds. Seeds are sown in nursery beds or plug trays, and seedlings are transplanted to the main field when 4 to 6 weeks old. Plant spacing is typically 30 to 45 cm between plants and 45 to 60 cm between rows for African marigold. Pinching the apical bud 2 to 3 weeks after transplanting promotes branching and higher flower yield.

Indian commercial cultivation practices: First planting is done in July-August for October-November festival season. Second planting in September-October for December-January wedding season. African marigold *Tagetes erecta* cv. 'Pusa Narangi Gaiinda' is among the most widely grown varieties in India. The crop matures within 70 to 90 days of transplanting. A well-maintained crop yields 20 to 30 tonnes of fresh flowers per hectare per season.

Uses and Applications

Religious and ceremonial: The primary use in India — garlands, decorations, loose flower offerings, and festive ornamentation. Used across all Hindu, Sikh, and some Muslim celebrations.

Natural pest repellent: Marigold roots secrete a compound (thiopene) that repels root-knot nematodes in soil. It is planted as a companion crop alongside tomatoes, chillies, and other vegetables in integrated pest management systems.

Natural food colourant and nutraceutical: Marigold oleoresin contains lutein and zeaxanthin, carotenoid pigments that are added to poultry feed and are sold as eye health supplements in human nutrition products.

Medicinal: Used in folk medicine for wound healing, skin inflammation, and eye disorders. Extracts of *Tagetes* species have demonstrated antimicrobial and antifungal activity.

Insect repellent: The scent of marigold foliage repels aphids, whiteflies, and other garden pests when planted as a border crop.

Interesting Facts

- India is the world's largest consumer of marigold flowers, driven almost entirely by religious and festive demand.
- During the Dussehra and Diwali season, a single marigold grower near Pune or Bengaluru can earn several lakh rupees in just 2 to 3 weeks.
- Marigold leaves contain pyrethrin-like compounds and have been used as a traditional mosquito repellent in rural India — branches are burned to drive away insects.
- The word 'marigold' is derived from 'Mary's Gold', a reference to the Virgin Mary in European Christian tradition.
- The Aztecs of Mexico considered marigold a sacred flower and offered it to their solar deity. The Spanish brought it to Europe in the 16th century, from where it spread to India.

Source: Flower AI Knowledge Base — MARIGOLD entry. Scientific name: *Tagetes erecta* (African Marigold) / *Tagetes patula* (French Marigold). Family: Asteraceae (Daisy family).